

SHIVAM KUMAR

+91 7482975394 | shivamk38812@gmail.com | Vaishali, Bihar | LinkedIn | GitHub

## PROFESSIONAL SUMMARY

QA Intern and Computer Science Engineering student with hands-on experience in manual testing, API testing, bug reporting, Jira workflows, and Playwright test automation. Also skilled in full-stack MERN development, REST API design, authentication, role-based access control (RBAC), transactional data consistency, and AI-powered document (RAG) workflows. Strong foundation in C, C++, Java, JavaScript, TypeScript, Node.js, React.js, MongoDB, MySQL, PostgreSQL, Git, and GitHub, with a quality-first, detail-oriented approach to software testing and development.

## EXPERIENCE

QA Intern | Toba Tech Solution May 2026 - Present

Performed manual testing of web application features to verify functionality, UI flow, usability, and expected user behavior.

Tested REST APIs using request/response validation, status code checks, payload verification, and error scenario coverage.

Explored Playwright for automation testing, including test script structure, browser actions, selectors, and end-to-end test flow design.

Used Jira to track bugs, update task status, document issues clearly, and coordinate QA feedback with the development team.

Prepared test cases, reported defects with clear reproduction steps, and supported regression testing after fixes and feature updates.

## EDUCATION

Bachelor of Technology in Computer Science and Engineering July 2023 - Present

Geetanjali Institute of Technical Studies, Udaipur | CGPA: 8.15

12th Grade (CBSE) 2022

Indraprastha International Secondary School, Muzaffarpur | Percentage: 68.2%

10th Grade (CBSE) 2020

Kendriya Vidyalaya Gannipur School, Muzaffarpur | Percentage: 66.2%

## PROJECTS

Fault-Tolerant Payment Ledger | Node.js, Fastify, TypeScript, PostgreSQL, React 19, Vite, Docker

Designed and built a fault-tolerant money transfer REST API focused on correctness under retries and concurrent requests.

Implemented idempotent transfers using durable Idempotency-Key storage, SHA-256 request fingerprints, safe replay, and 409 Conflict responses when the same key is reused with a different payload.

Ensured concurrency safety using SELECT...FOR UPDATE row locking, lexical lock ordering to avoid deadlocks, and atomic debit/credit plus audit-record writes within a single PostgreSQL transaction.

Built a React 19 + Vite ledger console (dashboard, transfer, activity, accounts) integrated with the REST API, with paginated account activity and structured problem+json error responses.

Packaged the application for deployment via Docker Compose, Render (API + DB), and Vercel/Netlify (UI).

HealthSync - Full-Stack Healthcare Platform | React.js, Node.js, Express.js, MongoDB

Built a healthcare platform with authentication and role-based access control (RBAC) for admin and patient users.

Implemented appointment booking, cancellation, and doctor availability management workflows.

Integrated real-time chat and video consultation features for remote patient-doctor interaction.

Added image and PDF file-sharing support for medical reports, prescriptions, and consultation records.

DocuGenius - AI PDF Chat Application | React.js, Node.js, Express.js, MongoDB, Pinecone, LLM APIs

Developed a full-stack MERN application that allows users to chat with PDF documents using AI.

Implemented Retrieval-Augmented Generation (RAG) for context-aware Q&A; by extracting, chunking, embedding, and retrieving PDF content.

Stored embeddings in Pinecone for fast similarity search and connected LLM APIs to generate relevant, context-grounded answers.

#### TECHNICAL SKILLS

Programming Languages: C, C++, Java, JavaScript, TypeScript, HTML, CSS

Frameworks / Libraries: React.js, React 19, Vite, Node.js, Express.js, Fastify, Playwright (Exploring)

Testing / QA: Manual Testing, API Testing, Test Case Design, Bug Reporting, Regression Testing, Jira

Databases / Cloud: MongoDB, MySQL, PostgreSQL, Pinecone, Docker, Render, Vercel, Netlify

Tools: VS Code, Git, GitHub, Jira, Docker Compose

Core Concepts: REST APIs, OOP, Authentication, Role-Based Access Control (RBAC), Idempotency, Concurrency Control, Database Transactions, RAG, Embeddings

Soft Skills: Teamwork, Problem Solving, Time Management, Communication

#### STRENGTHS

Manual and API testing mindset | Clear bug documentation | Full-stack project execution | Clean API integration | Concurrency-safe backend design | Logical problem solving | Fast learner with collaborative mindset